



INDIAN INSTITUTE OF INFORMATION TECHNOLOGY SONEPAT

भारतीय सूचना प्रौद्योगिकी संस्थान सोनीपत

Email: sonepatiit@gmail.com, website: www.iiitsonapat.ac.in

Mid Semester

Subject: COIT 303

Roll no. 12312021

Branch: IT, III-SEM

M.M.-15 marks

Time- 1 Hours.

Note: All questions are compulsory.

Q1	In a system with 4 GB RAM and a 64 KB direct-mapped cache with 64-byte blocks, determine the number of bits required for the tag, index, and block offset fields.	(2 marks)
Q2	A system with 8 MB RAM and a 4 KB direct-mapped cache has 64-byte blocks. The processor uses a 24-bit address space. Calculate the number of bits required for the tag, index, and block offset.	(2 marks)
Q3	In a system with 128 MB RAM and a 64 KB direct-mapped cache with 32-byte blocks, calculate the number of bits required for the tag, index, and block offset fields.	(2 marks)
Q4	Calculate Booth's Multiplication of +5 (0101) and -3 (1101)	(2 marks)
Q5	Calculate Floating Point Multiplication of 1.75 and 3.25	(2 marks)
Q6	In an independent memory system, the cache hit rate is 95%, the cache access time is 2 ns, and the main memory access time is 100 ns. What is the average memory access time?	(2 marks)
Q7	In a three-level cache system, the following data is available: - L1 cache hit rate = 90%, L1 access time = 2 ns - L2 cache hit rate = 85%, L2 access time = 8 ns - L3 cache hit rate = 95%, L3 access time = 25 ns - Main memory access time = 150 ns What is the average memory access time?	(2 marks)
Q8	Discuss Cache Mapping and its types.	(1 marks)



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Email: soncpatiiiit@gmail.com, website: www.iiitsoncpat.ac.in

Mid Sem-II

Subject: Design and Analysis Algorithm (UIT302)

Roll no 12312021

Branch: IT
(3rd SEM)

M.M.-15 marks

Time- 60 mint.

Note: All questions are compulsory.

Marks

Q-1: (a) Describe about Asymptotic notations. Define Master Theorem.

2+2

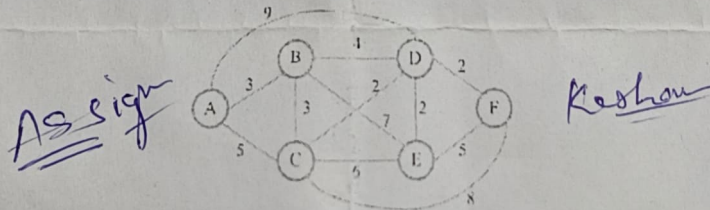
(b) "A stable sorting algorithm is one that preserves the relative order of equal elements in the input". Justify your answer with suitable example.

(bubble), Merge, Quick

Q-2: (a) Define the basic principal of Greedy approach.

1+3

(b) Solve the problem using Dijkstra's Algorithm to find the shortest path from vertex 'A' to all of the following weighted graph.



Q-3: (a) What is Backtracking Approach? List out two advantage of Backtracking approach?

1+3

(b) Considering the 4-Queens problem and provide two possible solutions to this problem with suitable example using backtracking technique.

Q-4 (a) Solved the following assignment problem using branch and bound technique.

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Email: sonapatiit@gmail.com, website: www.iiitsonepat.ac.in

Discrete Mathematics (^{ITC} ~~CSC~~ 301)

(B.Tech. MII) Semester II Examination)

Date: October 23, 2024

123/2021

Timing: 9:30 AM to 10:30 AM

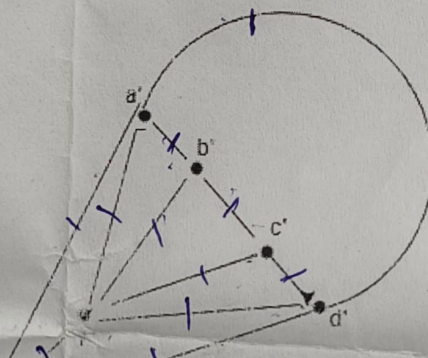
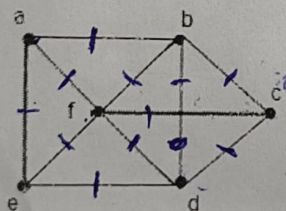
^{IT} ~~CSC~~ 2nd year (3rd semester)

Max mark: 15

All questions are compulsory

5 × 3 = 15 marks

1. Describe the Tower of Hanoi Problem and construct its recurrence relation then solve it?
2. Solve the recurrence relation $a_n - 5a_{n-1} + 8a_{n-2} - 4a_{n-3} = 3^n$, Using the method of Characteristic Roots.
3. Solve the recurrence relation $a_n - 2a_{n-1} + a_{n-2} = 2^{n-2}$, for $n \geq 2$ where $a_0 = 2$ and $a_1 = 1$. Using generating functions.
4. Define the following graphs with examples
 i) Regular Graph ii) Complete graph iii) Planer Graph iv) Bipartite Graph
5. Define Isomorphism of the graph. The following pair of non directed graphs are isomorphic? Justify your answer.





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Mid Sem-II

Subject Name: DBMS

Subject code: UIT306

Roll no 12312021

Branch: IT

3th Semester

M.M.-15 marks

Time- 60 min.

Note: (1) Attempt all the questions.

(2) All questions carry equal marks (3*5=15)

Q-1	For the given set of functional dependencies $F = \{A \rightarrow BC, B \rightarrow A, C \rightarrow D, D \rightarrow E, AB \rightarrow C\}$	(5 marks)
i)	Find all Super keys	
ii)	Find all candidate keys	
iii)	Is the relation in Boyce-Codd Normal Form (BCNF)? Justify your answer	
iv)	Is the relation in 2 NF? Justify your answer	
v)	Identify and remove any redundant functional dependencies from F.	
Q-2	Define serializability and explain the difference between serial and non-serial schedules. Provide examples of conflict serializable and view serializable schedules.	(5 marks)
Q-3	What is serializability in the context of transactions? Differentiate between conflict serializability and view serializability. Let $R_i(z)$ and $W_i(z)$ denote read and write operations on a data element z by a transaction T_i , respectively. Consider the schedule S with four transactions. $S: R_4(x) R_2(x) R_3(x) R_1(y) W_1(y) W_2(x) W_3(y) R_4(y)$ Is the given schedule conflict serializable? If yes, provide the equivalent serial schedule.	(5 marks)

End



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Roll no. 12312021

Time- 60 mint.

Mid Sem-II

Branch: IT

Subject: AFL (UIT-305)

M.M.-15 marks

Note: All questions are compulsory.

Q-1	Write down production rule and regular expression of following languages: (a) Language L over $\Sigma(a, b)$ such that each string having at least two length. (b) $L = \{a^n b^n c^m d^m, m, n \geq 1\}$ (c) $L = \{a^n b^{m+n} c^m, m, n \geq 1\}$ (d) Language L over $\Sigma(a, b)$ such that each string start and end with different symbol. (e) $L = \{ww^R \cup waw^R \cup wbw^R, \text{ such that } w \in (a,b)^*\}$	$(1.5+1.5+1.5+1.5+1.5)=7.5$
Q-2	Design PDA/NPDA of the following Languages: (a) $L = \{a^n b^{2n+1}, n \geq 1\}$ (b) $L = \{ww^R \mid w \in (a,b)^*\}$ and draw recursion tree for string: "aaaa". (c) $L = \{a^n b^n \mid n \geq 1\} \cup \{a^n b^{2n} \mid n \geq 1\}$ and draw recursion tree for string: "aabb".	$(1.5+1.5+1.5)=4.5$
Q-3	Design Turing Machine (TM) of following Languages: (a) Turing machine as Multiplier of two unary number. (b) Turing machine for the language: $L = \{wcw \mid w \in (a,b)^*\}$	$(1.5+1.5)=3$



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Mid Sem-II

Subject: Object Oriented Programming using JAVA (UIT304)

Roll no. 12312021

Branch: IT
(3rd SEM)

Maximum Marks: 15

Time: 60 minutes

Note: All questions are compulsory.

Marks

Q-1: (a)

Write a Java program to simulate a simple *file management* system. The program should be able to:

2

- Create a text file.
- Write data into the file using `PrintWriter`.
- Read data from the file using `Scanner`.
- Include exception handling for file operations such as `FileNotFoundException`

(b)

What is *exception handling* in Java? Differentiate between *checked* and *unchecked* exceptions. How would you design a custom exception class in Java? Include relevant code snippets.

2

Q-2: (a)

Explain the principles of inheritance and polymorphism in Java. How are they beneficial in object-oriented programming? Illustrate with examples involving both a base and a derived class. (10 Marks)

2

(b)

Describe the concept of abstract classes and interfaces. How do they differ? Provide an example demonstrating the use of an interface in Java.

2

Q-3:

Design an interface Movable with methods moveUp(), moveDown(), moveLeft(), and moveRight(). Create an abstract class Shape that implements Movable and add properties common to all shapes (like x and y coordinates). Now, design two classes Circle and Rectangle that extend Shape. Implement the interface methods in these subclasses and show how objects of these classes can be moved on a 2D grid. Ensure that the Circle and Rectangle classes handle the movement logic differently. Provide the implementation and demonstrate its use in a main method.

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Q-4:

Develop an application for managing a group of students. Create a Student class with fields like name, rollNumber, and GPA. Implement a method to sort the students based on GPA using an ArrayList of Student objects. Additionally, handle operations like adding new students, searching for students by roll number, and updating their GPA. Demonstrate how to handle these tasks efficiently with Java's Collections framework.

2

Q-5:

Explain the concept of object cloning in Java. How is the Cloneable interface used, and what is the role of the clone() method?

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